

JAYLEN WANG

Carnegie Mellon University

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Department of Electrical and Computer Engineering

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BRIEF BIOGRAPHY

My work bridges computer architecture and software systems, demonstrating the importance of that bridge in enabling sustainable data center systems via solutions that span the compute stack.

As the demand for web services continues to grow, data centers are scaling up to meet the demand, consuming a massive amount of energy and producing significant carbon emissions. **My research addresses the growing carbon emissions of data centers by analyzing inefficiencies across computer architecture and software systems and designing solutions to make these systems more energy and carbon efficient.**

My work is one of the first to examine the environmental impact of hyperscale web systems and to provide actionable insights to reduce it. My work integrates carbon efficiency into computer system design, as it is crucial for sustainable growth and access to critical web services in both developed and developing nations. Critically, my carbon-efficient solutions are starting to be deployed at hyperscale and are estimated to reduce global carbon emissions by $\sim 0.2\%$ by 2030—equivalent to eliminating the annual emissions of entire countries like Qatar/Venezuela.

My research has been recognized with an IEEE Micro Top Pick, an NSF Graduate Research Fellowship Program (GRFP) Award, the 2023 Benjamin Garver Lamme/Westinghouse Graduate Fellowship, a 2023 Ford Foundation Predoctoral Fellowship Honorable Mention, and the Jack and Mildred Bowers Scholarship in Engineering.

EDUCATION

Ph.D., Electrical and Computer Engineering

Advisor: Prof. Akshitha Sriraman

NSF GRFP Fellow

Dissertation Title: Enabling Sustainable Data Centers

Carnegie Mellon University

Aug 2022 – Present

B.Sc., Electrical Engineering

Research advisors: Profs. David Brooks & Gu-Yeon Wei

Minor in Computer Science

Graduated *Summa Cum Laude* (GPA: 4.0/4.0); *Member of Phi Beta Kappa*

Harvard University

Aug 2018 – May 2022

AWARDS AND HONORS

GreenSKU selected as an IEEE Micro Top Pick

Awarded to the top 12 computer architecture papers of 2024

2025

Finalist (2nd place) at SOSP Student Research Competition (SRC), Graduate Category

At SOSP'24 SRC, selected as finalist after poster session, and 2nd place after presenting

2024

Jack and Mildred Bowers Scholarship in Engineering

Full tuition support for third year of PhD

2024

NSF Graduate Research Fellowship Program (GRFP) Award Winner

Awarded \$171,000 as part of prestigious fellowship supporting exceptional graduate students in STEM

2023

Ford Foundation Predoctoral Fellowship Honorable Mention

Honorable mention given to top Ford Fellowship candidates

2023

Benjamin Garver Lamme/Westinghouse Graduate Fellowship

Full tuition support for second year of PhD

2023

Carnegie Institute of Technology Dean's Fellow

Awarded \$83,000 towards tuition, stipend, and travel

2022

Harvard SEAS Dean's Engineering Design Award

Awarded \$500 for having one of the top 7 (out of 43) best Senior engineering design projects

2022

Sophia Freund Prize	2022
Awarded \$1000 as highest ranking undergraduate in the Electrical Engineering department	
Phi Beta Kappa Member	2022
Admitted into Harvard's chapter, one of 146 out of 1962 (7.4%) students	
Derek Bok Center Distinction in Teaching	2020, 2021
Awarded to most highly rated (by students) TAs; received distinction in three semesters	
John Harvard Scholar	2020
Award given to top 5% (4.0 GPA) of students in respective class	
Harvard College Research Program Funding Recipient	2019
Awarded \$3,500 to perform independent research during the summer	
Detur Prize Winner	2019
Recognizes students with top academic standing in their first year at Harvard	

PEER-REVIEWED CONFERENCE AND JOURNAL PUBLICATIONS

- **Jaylen Wang**, Melissa Z. Pan, Udit Gupta, and Akshitha Sriraman. *(Under submission) Title omitted for anonymity*. International Symposium on High-Performance Computer Architecture (**HPCA 2027**).
- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvine, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warriar, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Enabling Sustainable Cloud Computing with Low-Carbon Server Design*. IEEE Micro. Issue: **Top Picks** in Computer Architecture from Conferences in 2024. Jun 2025.

Shows how, for the first time, to design and build real production-ready, low-carbon servers and how to evaluate servers' carbon-saving potential at scale prior to deployment.

- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvine, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warriar, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Designing Cloud Servers for Lower Carbon*. 51st International Symposium on Computer Architecture (**ISCA 2024**). Jun 2024. 

IEEE Micro Top Picks

Received the Artifact “Available”, “Functional”, and “Reproducible” ACM badges

First work to systematically show how to design cloud servers to significantly reduce cloud carbon emissions while meeting performance goals; this solution is being adopted across Microsoft Azure, with estimated savings of ~0.2% of global carbon emissions—on par with eliminating Qatar/Venezuela's annual emissions

PEER-REVIEWED WORKSHOP PUBLICATIONS & POSTERS

- **Jaylen Wang**, Fiodar Kazhamiaka, Pulkit Misra, Daniel S. Berger, and Akshitha Sriraman. *Safe CPU Oversubscription: One Size Does Not Fit All*. 3rd Workshop on Hot Topics in System Infrastructure (**HotInfra 2026**) held in conjunction with **ISCA 2026**. Jun 2026.
- Abnash Bassi, **Jaylen Wang**, Fiodar Kazhamiaka, Daniel S. Berger, and Akshitha Sriraman. *Store or Recompute? Characterizing the Carbon Tradeoff of KV Cache Retention in LLM Inference*. 11th Workshop on Energy Efficient Machine Learning and Cognitive Computing (**EMC2 2026**). Mar 2026.
- **Jaylen Wang**, Melissa Pan, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyper-scale*. Presented Poster at **SOSP 2024**. Nov 2024.
- **Jaylen Wang**, Asser Tantawi, Olivier Tardieu, and Akshitha Sriraman. *Making Multi-Cluster Scheduling Carbon-Aware*. 2nd Doctoral Workshop (**SysDW 2024**) held in conjunction with **SOSP 2024**. Nov 2024.
- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvine, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warriar, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Designing Cloud Servers for Lower Carbon*. 2nd Workshop on Hot Topics in System Infrastructure (**HotInfra 2024**) held in conjunction with **SOSP**. Nov 2024.
- **Jaylen Wang**, Melissa Pan, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyper-scale*. 6th Young Architect Workshop (**YArch 2024**) held in conjunction with **ASPLOS**. Jun 2024.

- **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Peeling Back the Carbon Curtain: Carbon Optimization Challenges in Cloud Computing*. 2nd Workshop on Sustainable Computer Systems (**HotCarbon 2023**). Jul 2023.  
Performs the first ever characterization of server generations for microservice-based web services to enable hardware lifetime extension
- Jialun Lyu, **Jaylen Wang**, Kali Frost, Chaojie Zhang, Celine Irvine, Esha Choukse, Rodrigo Fonseca, Ricardo Bianchini, Fiodar Kazhamiaka, and Daniel S. Berger. *Myths and Misconceptions Around Reducing Embedded Carbon for Cloud Platforms*. 2nd Workshop on Sustainable Computer Systems (**HotCarbon 2023**). Jul 2023.  
- **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyperscale*. 1st Workshop on Hot Topics in System Infrastructure (**HotInfra 2023**) held in conjunction with **ISCA**. Jun 2023. 
- **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Characterizing Datacenter Server Generations for Lifetime Extension and Carbon Reduction*. 1st Workshop on NetZero Carbon Computing (**NetZero 2023**) held in conjunction with **HPCA**. Feb 2023. 
- Sahana Rangarajan, **Jaylen Wang**, Sara Mahdizadeh Shahri, Pratyush Patel, and Akshitha Sriraman. *Designing Equitable Data Center Scheduling Systems*. Career Workshop for Inclusion and Diversity in Computer Architecture (**CWIDCA 2022**) held in conjunction with **MICRO**. Oct 2022. 
- **Jaylen Wang**, Abdulrahman Mahmoud, Gu-Yeon Wei, and David Brooks. *A Dataflow-Aware Fault Resilience Analysis Framework for Deep Neural Network Accelerators*. 4th Young Architect Workshop (**YArch 2021**) held in conjunction with **ASPLOS**. Mar 2022. 
Introduces a new framework and tool to quickly and accurately assess the reliability of deep neural network accelerator designs to random bit flips, providing insights for resilient accelerator design

INVITED SEMINAR TALKS

Designing Cloud Servers for Lower Carbon

	University of Washington (CSE), Host: Prof. Baris Kasikci	Jun 2025
	MSR India, Host: Dr. Ramachandran Ramjee	Oct 2024
	AMD Research, Host: Dr. Srilatha (Bobbie) Manne	Oct 2024
	Intel Processor Architecture Research Lab, Host: Dr. Shankar Balachandran	Oct 2024
	Princeton University (CS), Host: Prof. Margaret Martonosi	Oct 2024
	Rutgers University (CS), Host: Prof. Santosh Nagarakatte	Oct 2024
	Systems Research @ Google (SRG) Group, Host: Dr. David Culler	Sep 2024
	ISCA, Buenos Aires	Jun 2024
	University of California San Diego (CSE), Host: Prof. Jishen Zhao	Apr 2024
	University of California Riverside (ECE), Host: Prof. Shaolei Ren	Apr 2024
	University of Southern California (ECE), Host: Prof. Murali Annavaram	Apr 2024

SELECTED PRESS

- *IEEE Spectrum* Nov 2024
“Servers Get a Second Life for Sustainability” 
- *MSN* Oct 2024
“Researchers discover new method to curb cloud computing’s harmful impact: ‘We targeted planned obsolescence’” 
- *TechXplore* Oct 2024
“New methodology enables design of cloud servers for lower carbon” 

- *Carnegie Mellon University News* “Enabling Sustainable Cloud Computing with Low-Carbon Server Design”  Aug 2024
- *Carnegie Mellon University News* “Designing Cloud Servers for Lower Carbon Emissions”  May 2024

AWARDED GRANT PROPOSALS COLLABORATED ON

- *AWS Cloud Credit for Research*, “Carbon-Aware Scheduling to Reduce Hyperscale Carbon Emissions”, Award: \$76,000, Award period: 2024–25

PROFESSIONAL SERVICE

Program committee chair

- *Co-Chair* for Wild and Crazy Ideas (WACI 2026) held in conjunction with ASPLOS Mar 2026

Organizer/founder of workshop or panel

- *Workshop Lead Organizer* for 2nd Workshop on Ethical Systems and Architecture Design (HotEthics 2026) held in conjunction with ASPLOS  Mar 2026
- *Workshop Co-Founder* for 1st Workshop on Ethical Systems and Architecture Design (HotEthics 2024) held in conjunction with ASPLOS  Apr 2024
- *Panel Co-Organizer* for Panel on Sustainable Systems at The 19th Workshop on Hot Topics in Operating Systems (HotOS 2023)  Feb 2023

Invited contributor, speaker, or panelist

- *Invited Speaker* for ACT Tutorial (Architectural Carbon Modeling Tool) held in conjunction with MICRO Nov 2024
- *Graduate Student Panelist* for Undergrad Architecture Mentoring Workshop (uArch) held in conjunction with ISCA Jun 2024
- *Co-Author* in ACM SIGARCH Computer Architecture Today blog article *Reducing Embodied Carbon is Important* (ACM SIGARCH 2023)  Aug 2023

Artifact evaluation committee member

- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025) Feb 2025
- International Symposium on Microarchitecture (MICRO 2024) Sep 2024
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2024) Oct 2023

PROFESSIONAL EXPERIENCE

PhD Software Engineering Intern, Google, Durham, NC May 2026 – Aug 2026
continued part-time as Student Researcher Aug 2026 – Dec 2026

Mentors: Jixin Feng and Soheil Khadir

Leading a project on reducing the power consumption of host servers that support large-scale ML training workloads.

Azure Systems Research Intern, Microsoft Research, Redmond, WA May 2025 – Aug 2025
 Mentors: Fiodar Kazhamiaka and Daniel Berger

Led research project on novel approaches to optimize data center carbon efficiency through a combination of strategic resource management and workload identification; findings have influenced oversubscription and server-design strategies at Microsoft Azure.

IBM Research Intern, IBM Research, Yorktown Heights, NY May 2024 – Aug 2024
 Mentor: Asser Tantawi

Created new frameworks and strategies for more sustainable cross-cluster batch job scheduling.

Azure Systems Research Intern, Microsoft Research, Redmond, WA
Mentors: Fiodar Kazhamiaka and Daniel Berger

May 2023 – Aug 2023

Designed and built a novel framework to understand the data center-scale impacts of server design to improve resource and carbon efficiency. Work has reached early production and is being considered by Azure as a strategy towards achieving Microsoft’s 2030 decarbonization targets.

Graduate Research Assistant, Carnegie Mellon University
Advisor: Prof. Akshitha Sriraman

Aug 2022 – Present

Introducing sustainability as a first-order hardware/software system design metric for hyperscale systems and redesigning data center systems to be carbon-efficient.

Undergraduate Research Assistant, Harvard University
Lab: Harvard Architecture, Circuits, and Compilers Group
Advisors: Dr. Abdulrahman Mahmoud, Profs. Gu-Yeon Wei and David Brooks

Mar 2021 – Aug 2022

Developed a hardware-aware framework for analyzing the resilience of deep neural network accelerators to soft errors, considering the reuse of values in an accelerator’s dataflow propagation.

Engineering Development Group Intern, MathWorks
Team: Deep Learning HDL Toolbox
Supervisors: Wang Chen, Siyuan Xu

May 2021 – Aug 2021

Enabled efficient mapping of non-square convolution kernels onto square processing-element arrays, allowing users to deploy models using non-square kernels onto FPGAs

Undergraduate Research Assistant, Harvard University
Lab: Harvard Edge Computing Lab
Advisor: Prof. Vijay Janapa Reddi

May 2020 – Aug 2020

Analyzed how using SLAM for localization affected efficiency and power usage within autonomous drone applications by integrating SLAM algorithms into an open-sourced drone benchmarking framework.

Undergraduate Research Assistant, Harvard University
Lab: Hoffman Physics Lab
Advisor: Prof. Jenny Hoffman

May 2019 – Aug 2019

Developed a tensioning system for an XY-walker system to extend the range of a scanning tunneling microscope used to research the proximity effect of superconductivity.

STUDENT MENTORING

PhD

- Abnash Bassi 2025–Present
First author on EMC2 2026 workshop paper

Master’s

- Aaron Hsiung (now at TSMC) 2025–2026
- Josiah Miggiani 2025
- Melissa Pan (now a PhD student at UC Berkeley) 2023–2024
Co-author on SOSF 2024 poster, YArch 2024 paper, and a paper under submission

Undergraduate

- Onyx Mayer 2025
- Andrew Gao (now at Vooma) 2024
- Yuxuan Zhu (now at Intuit) 2023

TEACHING EXPERIENCE

Graduate teaching assistant, Carnegie Mellon University

- *Computer Architecture and Systems (18-742)*; Prof. Phil Gibbons Spring 2026

Invited guest lecture on sustainable computing

- *Computer Systems & the Hardware-Software Interface (18-344)* Prof. Akshitha Sriraman & Brandon Lucia *Fall 2024*

Undergraduate teaching assistant, Harvard University

- *Systems Programming and Machine Organization (CS 61)*; Prof. Eddie Kohler *Fall 2020, 2021*
- *Circuits, Devices, and Transduction (ES 152)*; Profs. Gage Hills & Woodward Yang *Fall 2021*
- *Systems and Control (ES 150)*; Profs. Na Li & Yue Lu *Fall 2021*
- *Introduction to Electrical Engineering (ES 50)*; Profs. Chris Lombardo & Marko Loncar *Spring 2021*
- *Integration, Series and Differential Equations (MATH 1B)*; Dr. Hakim Walker *Fall 2019*

LEADERSHIP & VOLUNTEERING

- CMU Institute of Technology K-12 Outreach, STEM Volunteer *2023–Present*
- President of Harvard Club Tennis *2019–2022*
- President of Harvard College Engineering Society *2020–2021*
- Co-President of Harvard Engineering Peer Concentration Advisors *2020–2021*

REFERENCES

1. Prof. Akshitha Sriraman (akshitha@cmu.edu)
Assistant Professor, Carnegie Mellon University
2. Dr. Ricardo Bianchini (ricardob@microsoft.com)
Technical Fellow, Corporate Vice President, Microsoft Azure
3. Dr. Parthasarathy Ranganathan (parthas@google.com)
Vice President and Technical Fellow, Google
4. Prof. Babak Falsafi (babak.falsafi@epfl.ch)
Professor, École Polytechnique Fédérale de Lausanne (EPFL)
5. Prof. David Brooks (dbrooks@g.harvard.edu)
Haley Family Professor of Computer Science, Harvard University
6. Prof. Udit Gupta (ugupta@cornell.edu)
Assistant Professor, Cornell University